Team Project A

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Project name: Vehicle parking classification using OpenCV and Yolov5

1. Introduction

This project aims to develop a vehicle classification system that leverages deep learning and computer vision techniques. The core objective is to identify and classify vehicles such as cars, buses, motorcycles, and bicycles from video footage. By using pre-trained deep learning models and OpenCV, the system efficiently processes frames from a video stream to detect and label vehicles in real time.

2. Problem Statement

Manual vehicle monitoring systems are inefficient and not scalable for large-scale deployment. There is a growing demand for automated systems that can monitor traffic, detect vehicles, and classify them accurately. This project addresses this problem by developing an AI-based solution using a deep learning model for real-time vehicle classification.

3. Tools and Technologies Used

- Python
- OpenCV
- MobileNet SSD (Single Shot MultiBox Detector)
- Pre-trained Caffe model
- Jupyter Notebook

4. Methodology

The system uses the MobileNet SSD deep learning model trained on the Caffe framework. The workflow involves loading the model, preprocessing input frames, detecting objects, and classifying vehicle types. OpenCV is used to process the video input, draw bounding boxes, and annotate the detected vehicles in real time. Each frame of the input video is converted to a blob, passed to the network, and the detection output is used for labeling.

5. Implementation

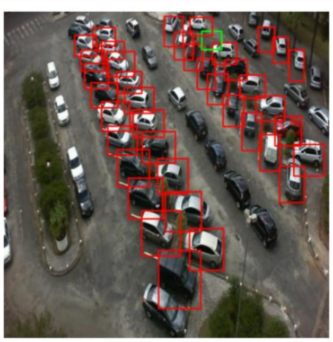
The implementation was done in Python using OpenCV and the MobileNetSSD Caffe model. The model was integrated into a script that processes each video frame and identifies objects. Vehicles such as cars, buses, and motorbikes are filtered from the list of detected objects, and bounding boxes are drawn around them with their labels. The annotated frames can be saved or displayed in real time, providing visual feedback.

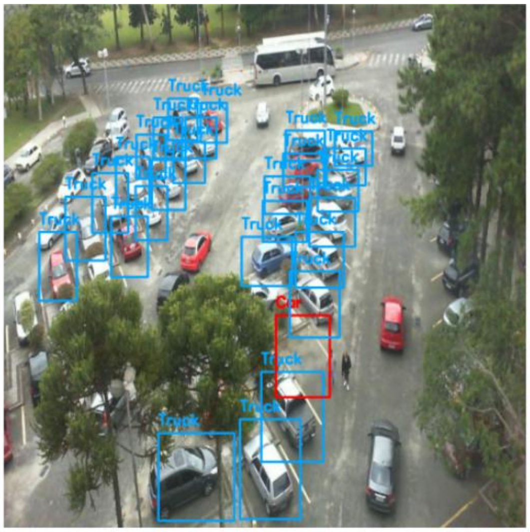
6. Results and Output

The system was tested on video footage containing a mix of vehicle types. It successfully identified and classified vehicles in most frames. The results indicate that MobileNet SSD is effective for real-time vehicle classification. Sample outputs show bounding boxes and labels on the detected vehicles, demonstrating the system's capabilities.

```
for i in range(5):
    aug_img = next(aug_iter)[0]
    img = array_to_img(aug_img)
    img.save(os.path.join(aug_save_path, f'augmented_{i+1}.jpg'))

    plt.subplot(1, 5, i+1)
    plt.imshow(img)
    plt.axis('off')

plt.tight_layout()
plt.show()
```

```
    plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
    plt.axis("off")
    plt.show()

    break
```





7. Conclusion

This project demonstrates the practical use of deep learning and computer vision for automated vehicle classification. The implemented system is capable of detecting and labeling multiple types of vehicles from video input in real time. Such systems can be beneficial for traffic monitoring, parking management, and smart city infrastructure.